A Data-Driven Detection System for Predicting Stress Levels from Autonomic Signals

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***Abstract*—This paper introduces and details a detection system for continuous monitoring of psychological stress. The formulation of classes relating to varying levels of stress intensity is described. This is necessary for determining the therapy needed to alleviate the associated effects, particularly in population groups suffering from chronic and mental illnesses such as diabetes and depression. The data-driven detection system mainly comprises kernel principal component analysis for dimensionality reduction, and nearest neighbour classifier for supervised learning to determine the associated stress intensity. We evaluate the generalised stress detection system using a 3-fold cross validation and a test set comprising an independent subject. We obtain a 0.66 F1-score with a precision of 0.70 and a recall of 0.67 over 4 classes of stress: no stress, low stress, moderate stress, and high stress.**

I. Introduction

Chronic stress is an increasingly concerning factor affecting a significant proportion of the population. According to a recent study administered in the United Kingdom by YouGov, 74% of the respondents reported being stressed to the point of being overwhelmed and unable to cope [1].

Stress can be defined as a heightened state induced by a set of adverse situations (stressors) [2]. This leads to disequilibrium in the regulation of bodily functions in order to facilitate an adequate response to the onset of the stressor [2], [3]. This heightened state is characterized by an increase in attention, brain function, and a subsequent suppression of physiologic systems such as metabolic and immune systems. Such a response is necessary for survival, however in the presence of a persistent stressor, the stress response turns chronic and maladaptive - posing significant negative effects on quality of life [2].

The autonomic nervous system is generally involved in maintaining homeostatic processes that maintain the set point for metabolic, cardiovascular, respiratory and endocrine systems among others [3], [4]. Physical and psychological stress leads to activation of the two main autonomic systems: sympathetic nervous system (SNS) and parasympathetic nervous system (PNS).

The SNS facilitates increased metabolic output. This is reflected in blood vessel constriction, a reduction in skin temperature, as well as an increase in blood pressure, blood glucose, and heart rate. On the other hand, the PNS is associated with the restoration of vital organ systems. Chrousos et al. [2] motivate the quantification of PNS activity can therefore be employed as an index of stress and stress vulnerability. At the onset of stress, the two centres in the brain are activated, primarily the hypothalamic pituitary adrenal (HPA) axis that releases cortisol and the SNS [3], [4]. The PNS activity is depressed in tandem with the activation of these two pathways [4].

In recent years, self-reported scales such as the Perceived Stress Scale (PSS), Self-Assessment Manikin (SAM) are used to provide feedback on stress [5]. However, this approach suffers due to the differences in stress vulnerability. In addition, most self reported data requires a considerable amount of attention in the context of ambulatory settings and thus not feasible for continuous monitoring.

A challenge in developing a robust stress detection system is taking into consideration the variability in personality and as a result differences in individual responses to various stressors. The availability of wearable sensors to measure physiological variables facilitate indexing stress. Consequently, there have been multiple approaches that aim to automatically detect stress.

The initial work in the detection of stress was limited to singular physiological signals [6] that enabled discrimination between the presence or absence of stress. The benefit in this approach is the ability to develop a low cost sensors with reduced computational demands. On the other hand, the complexity of organ system interactions means other activities such as exercise [7] can lead to similar activation of sections of the autonomic pathway thereby affecting the efficacy of application in a real world setting.

The main contribution of this work is a stress detection system that predicts 4 stress levels from different stressor types, accounting for habituation and repeated exposure as in real life. Recent works have taken a multimodal approach to improve robustness [8]. However, these works take into consideration a combination of physiological signals to mostly discriminate between two levels of stress or similar stressor types [6]. Our work facilitates an approach for incorporating stress detection in diabetes management which would improve glucose control [9], [10].

II. Methodology

A. Dataset

The dataset comprises 12 (6 male and 6 female) participants go through five sessions that are meant to elicit stress responses. The participants have an age range of 18-35 years. Figure 1 above shows a description of the sessions that the

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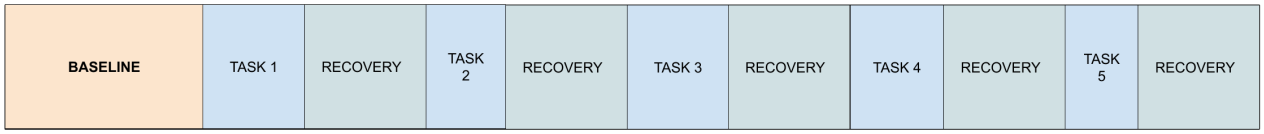

Fig. 1. A description of the stress session that each participant undertakes. Each session comprised 5 tasks that are separated with 15 minute recovery sessions to enable recovery and take self-reported assessments and saliva samples . An initial baseline measurement is also taken. Task 1 = IADS [11]; Task 2 = Stroop test [12]; Task 3 = NxN puzzle ; Task 4 = IADS [11]; Task 5 = Cold pressor test [13]

participant completes. In the period succeeding each task is a recovery period where we allow the participants vitals to return to the baseline levels.

We use an Empatica E4 and QardioArm blood pressure monitor to collect four physiological signals: blood volume pulse, electrodermal activity, skin temperature, and blood pressure.

B. Data preprocessing

As seen in Figure 1, on completion of each session the participant fills in a State-Trait Anxiety Inventory (STAI) A-State form [14]. This scale provides a quantified representation of the stress response of the participant. In addition, the A-Trait form that the participant completes during the baseline period quantifies the generalised propensity to anxiety prior to undertaking a single task. We perform k-means clustering of the responses for each task with all the participants in order to obtain four stress levels: no stress, low stress, moderate stress, and high stress. These labels are further validated by comparison with saliva cortisol samples.

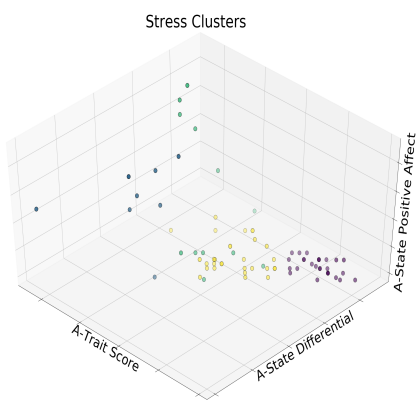

Fig. 2. A 3-dimensional plot showing the stress clusters based on A-Trait score, A-State differential score, and A-State positive affect score. The clusters are also determine on the absolute A-State score and the A-State negative affect score.

C. Feature Extraction

The physiological signals undergoes a number of preprocessing steps before the relevant features are extracted. First, a low-pass Butterworth filter is used to remove the noise from the EDA ($\geq$ 2Hz). In addition, the artifacts are removed using Ledalab v3.4.9 [15]. In order to accurately process the physiological signals to obtain the features, we use a 5 minute sliding window with 50% overlap. The features are then extracted from the physiological signals.

The physiological signals are processed and to obtain the necessary features for training the stress detection system. In order to obtain the most accurate features reflecting indications of stress, a 5 minute window is used to process the signals. This particularly pertinent for accurate analysis of very low frequency components (0.0033 Hz $\leq$ VLF $\leq$ 0.03 Hz) [16] for the heart rate variability. The frequency domain features are calculated using a Lomb-Scargle periodogram.

Blood volume pulse: Photoplethysmography (PPG) provides an optical measure of the blood volume pulse. Given this, it is possible to infer from the beat-beat variability. Baevsky et al. [16] show the various features that can be processed from the heart rate variability (HRV). This mainly covers features in the time-domain and the frequency domain.

Electrodermal activity (EDA): Skin conductance is defined by the electrical activity of a person due to secretion of sweat glands over the skin. In response to stressor onset, the HPA induces motor activity in the sweat glands. The skin conductance is decomposed into a phasic skin conductance response (SCR) and tonic skin conductance level through a process of nonnegative deconvolution. This is carried out using Ledalab v3.4.9 [15].

Skin temperature (ST): The skin temperature is recorded using a thermopile sensor. The activation of the autonomic nervous system in response to a stressor leads to a constriction of the blood vessels closer to the surface of the skin. This results in a decrease in skin temperature and can thus serve as an indicator of stress.

Blood pressure: The blood pressure values obtained from the QardioArm monitor provide non-continuous measures of the systolic and diastolic pressure values. The measurement period is doubled in order to obtain a more accurate value. Given the relatively slow changing nature of blood pressure, it is assumed that the value obtained is constant over the entire measurement period.

D. Detection system

The processed features are then standardized and used to train a detection system that comprises dimensionality reduction and supervised learning algorithm. As shown in Figure 3, the first stage of the detection system is a kernel

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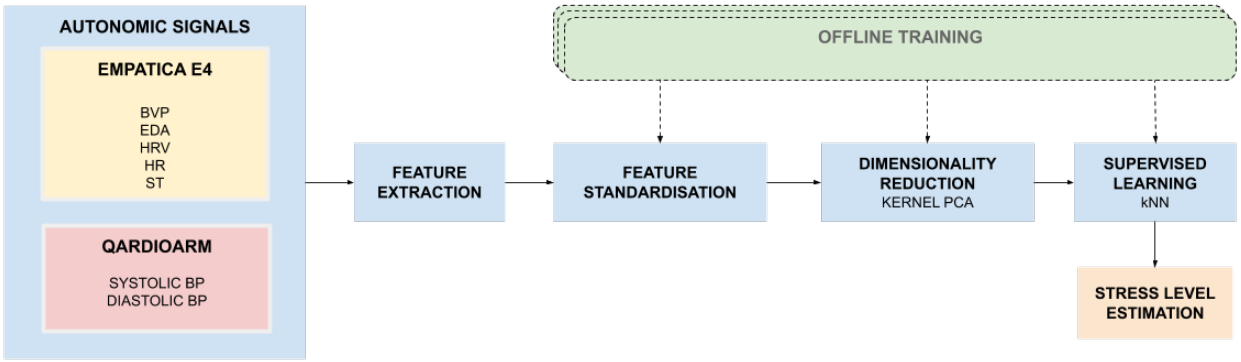

Fig. 3. A description of the stress detection system is shown. The autonomic signals are first sensed by the Empatica E4 and QardioArm BP Monitor. The relevant features are then extracted from these signals. The latter three stages require training data to which is done offline and evaluated with testing data.

principal component analysis (KPCA). PCA is employed as an intermediary stage before supervise learning to project the features from a higher dimensional mapping to a lower dimensional mapping. The kernel implemented to reduce the training data dimensions to the first *n* principal components is a Gaussian radial basis function. Although for the discrimination of binary stress/no stress states, in a multi-class setting where we discriminate between 4 states, the suggestion of habituation with stress responses motivates the possibility of a non-linear mapping between the input space and the associated number of classes.

III. Results

The kernel PCA dimension-reduced training set is then fed to a number of multiclass supervised learning algorithms. This is shown in Figure 3. The processed features are used to train a k-nearest neighbour (kNN) classifier (where k = 4). This is compared with the approach of Akmandor et. al [8] which consists of linear PCA and kNN.

The systems are evaluated based on the F1-score, Precision, and Recall. These metrics are detailed below:

$$Precision = \frac{TP}{(TP + FP)}$$
$$Recall = \frac{TP}{(TP + FN)}$$
$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

where TP, FP, FN denote true positive (correct prediction of stress level), false positive (wrong prediction of presence of stress), and false negative (wrong prediction of absence of stress) respectively. The precision score represents the fraction of correct stress level predictions relative to total positive predictions made. Recall, on the other hand, measures of the The F-score represents a harmonic mean of the recall and precision.

Table I shows the comparison between performance using PCA for dimensionality reduction using a radial basis function (RBF) as the kernel and linear PCA described in [8]. The results highlight the difference between binary and multiclass stress level classification. The average performance of our system over the cohort of participants is greater than (F1: +0.07; precision: +0.10; recall: +0.06) the performance of the system with a linear PCA. In addition, we note that the standard deviation of our system is lower for all considered metrics.

TABLE I
A comparison of the performance of different stress detection systems

Metric	Model	
	RBF PCA + kNN	LINEAR PCA + kNN
F1 Score	$\mathbf{0.66 \pm 0.14}$	0.59 ± 0.15
Precision	$\mathbf{0.70 \pm 0.14}$	0.60 ± 0.16
Recall	$\mathbf{0.67 \pm 0.12}$	0.61 ± 0.15

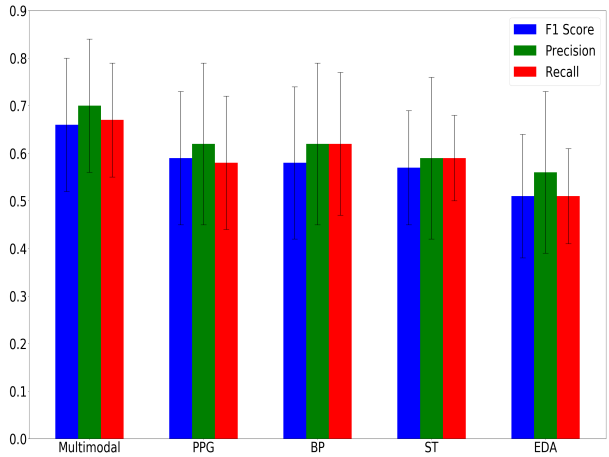

Fig. 4. A comparison of the stress detection system performance between multimodal approaches and various unimodal approaches.

The next experimental evaluation details the performance of our proposed system using multimodal approach relative to unimodal approach with the various physiological signals. Figure 4 shows a bar chart representation of the F1 score,

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precision, and recall of the detection system under the investigated approaches. The multimodal approach demonstrates the best approach with an improvement (F1 score: +0.07) over the best unimodal approach. The unimodal approach the best F1 score is shown to utilise the PPG signal and associated features. BP and ST provide the next best performance with 0.58 and 0.57 F1-scores respectively. The worst performing physiological signal was the EDA which had an F1-score of 0.51.

IV. Discussion

The results detailed in the section above highlight two salient observations.

Firstly, we posit that the reason for the increase in performance lies in the non-linear relationship between the multiple stress intensities and the effect on the physiological signals. This would be in contrast to the mostly linear relationship between the relaxed state and stress states. This could also explain the lower standard deviation observed in the results since the approximation of principal components to be linear may degrade performance depending on the degree of non-linearity observed between different classes for different subjects.

Inter-subject variability could account for the stronger performance in the multimodal approach over the various unimodal approaches. For each participant, different signals may exhibit varying sensitivities in response to stressors, leading to relatively more robust performance when using a multimodal approach.

Existing literature has shown that generalised models tend to have a lower accuracy relative to individualised models. However, the observation period necessary to gather the individual training data samples for multiple stress states may be prohibitive. Future work will investigate overcoming this limitation and improving performance by clustering participants in order to pool participants with more similar stress response traits.

In addition, a multimodal approach currently requires multiple wearable devices which can be cumbersome. One approach would be to incorporate estimates of blood pressure from PPG and ECG signals through pulse transit time (PTT) [17], enabling consolidation into one device.

V. Conclusion

Wearable devices offer an opportunity to provide continuous, ambulatory and non-invasive means of recording physiological signals that reliably reflect the activity of the autonomic nervous system - an indicator of stress. We describe a system that can be developed towards improving diabetes management. As the damaging effects of stress become more prevalent, it is necessary to reliably detect various levels of stress and provide the necessary therapy to relieve the associated effects.

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